

Emma Verma

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 78% (Strong Candidate)

emma.verma599@example.com | +1 (296) 457-1002 | Seattle, WA

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: MQTT, FreeRTOS, Embedded Linux, I2C/SPI/UART, PCB Debugging, Python

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core MQTT service, reducing latency by 58%.
- Led a squad of 4 engineers delivering scalable microservices processing 43M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 30%.

Associate Embedded Systems & IoT Firmware Engineer — *Intel* (2019 - 2023)

- Designed and maintained FreeRTOS data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using Embedded Linux saving \$31k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [MQTT, FreeRTOS, Embedded Linux, I2C/SPI/UART]

Engineered end-to-end production solution leveraging MQTT and FreeRTOS with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Embedded Linux, I2C/SPI/UART, PCB Debugging, Python]

Architected high-resilience system integrating Embedded Linux with 99.9% uptime.

EDUCATION & CREDENTIALS

M.Tech in Data Science & Artificial Intelligence

BITS Pilani — 2015 - 2019 (Score: 9.32 / 10.0)